

Uncovering Simultaneous Breakthroughs with a Robust Measure of Disruptiveness

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Same discovery, opposite scores?

Englert, F., & Brout, R. (1964)

$D = 0.15$
Top 2.8 %



Higgs, P.W. (1964)

$D = -0.27$
Bottom 0.1 %

Peter Higgs and Englert & Brout independently proposed the same theory at the same time and received 2013 Nobel Prize.

Yet the standard disruptiveness metric ranks **Englert** in the **top 3%** and **Higgs** in the **bottom 0.1%**, only because Higgs cite Englert.

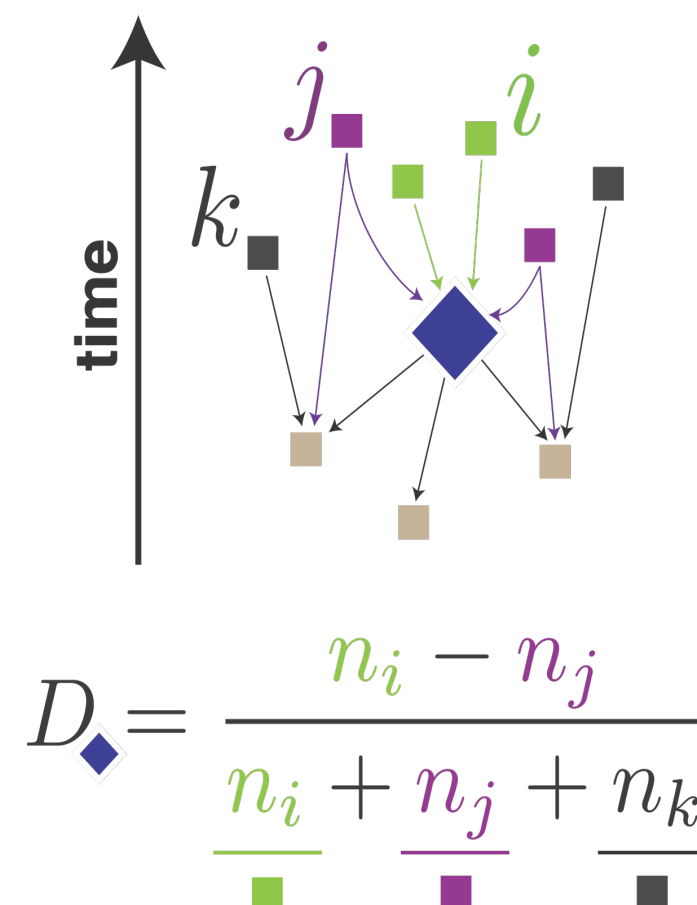
The Problem with Counting Citations

Existing measures quantify disruptiveness by counting how many future papers cite a focal work without also citing its references. But this simple counting approach:

Is **sensitive** — a single citation can flip the score

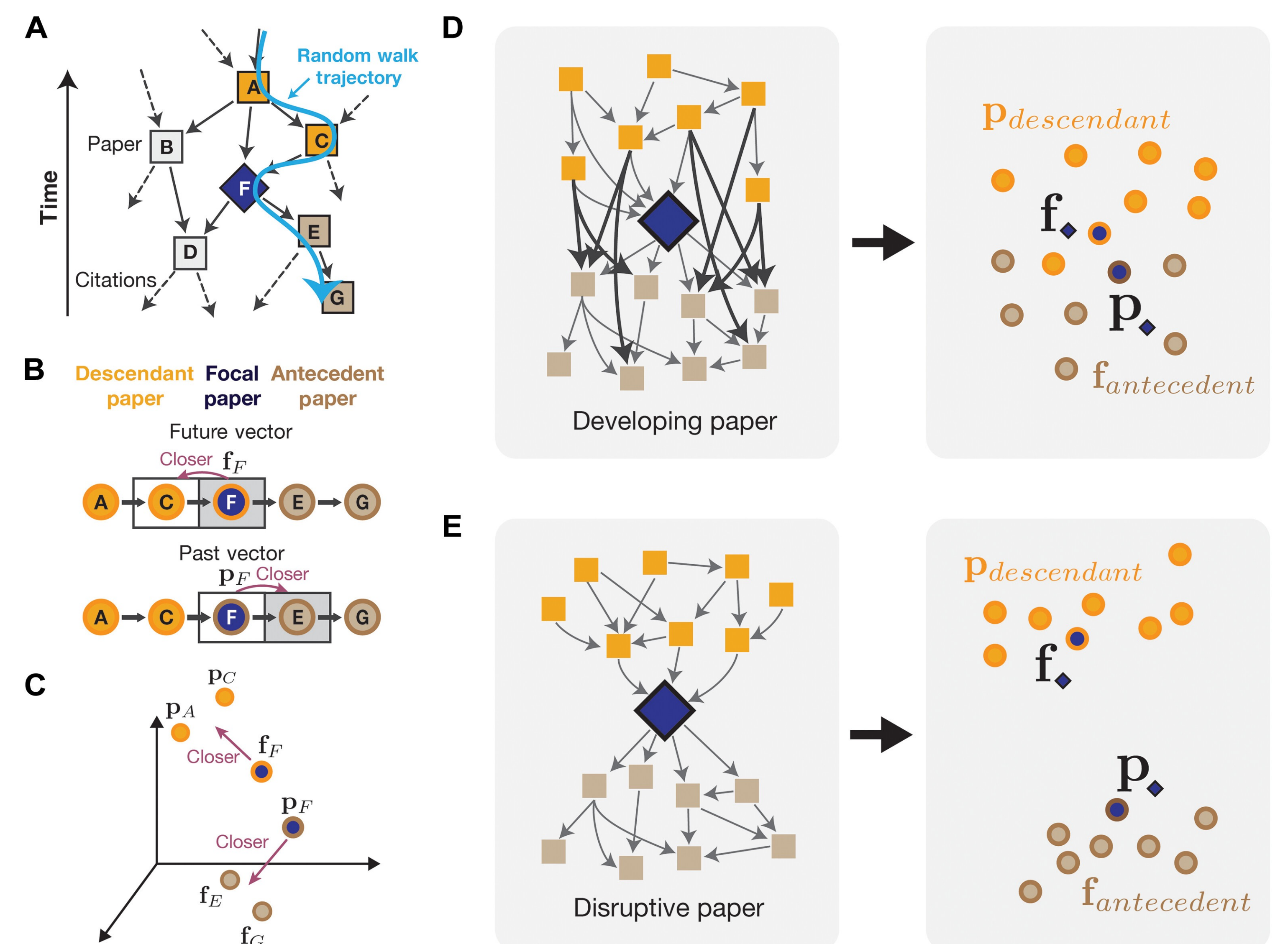
Only sees **immediate neighbors**, missing the bigger picture

Often **fails when two teams independently make the same breakthrough**



$$D_i = \frac{n_i - n_j}{n_i + n_j + n_k}$$

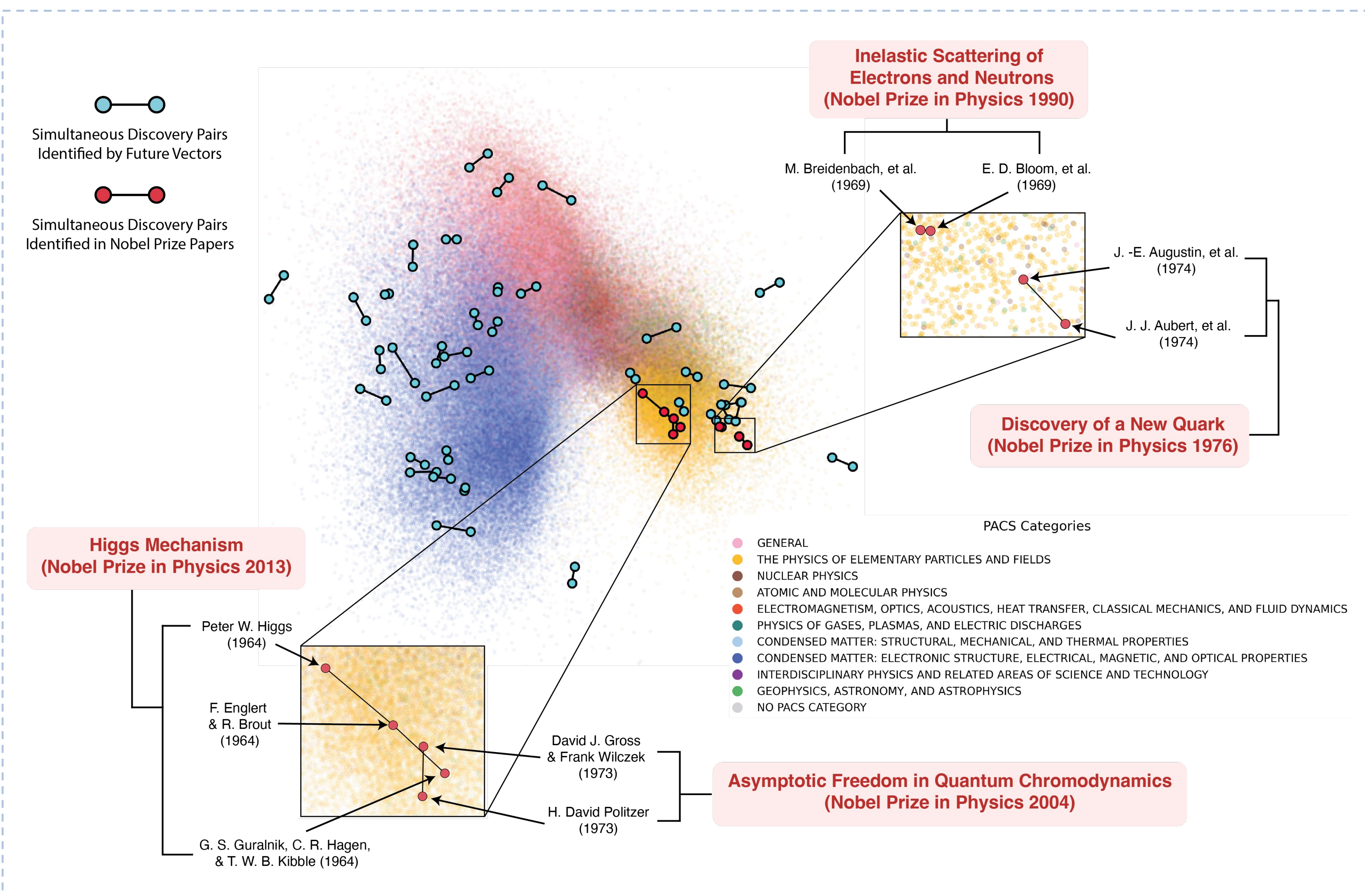
How can we reliably measure the transformative nature of scientific work?



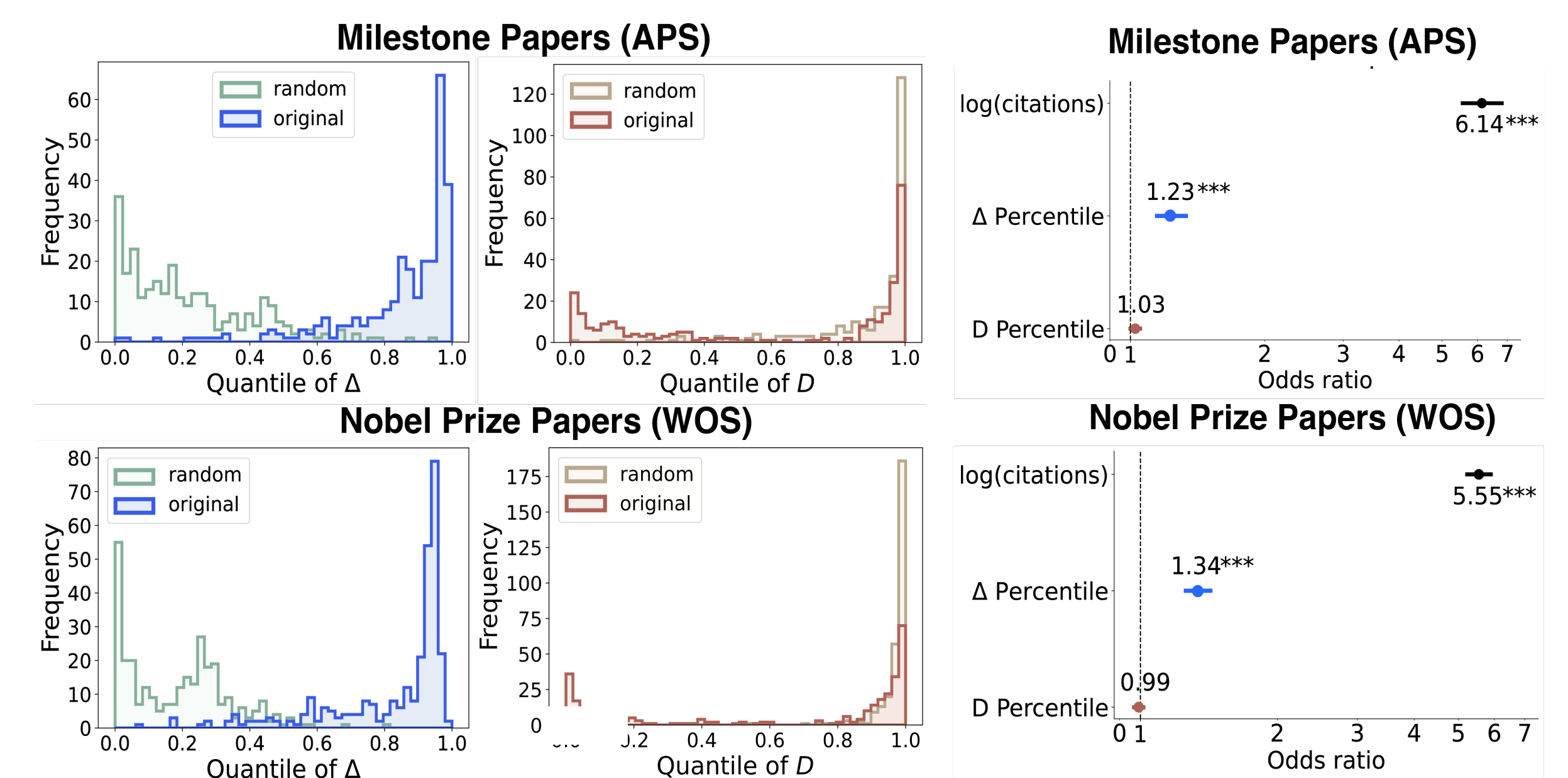
For each paper, learn two vectors in embedding space:
Past vector $\mathbf{p}_i \rightarrow$ references **Future vector** $\mathbf{f}_i \rightarrow$ citations

$$\Delta_i = 1 - \frac{\mathbf{f}_i \cdot \mathbf{p}_i}{\|\mathbf{f}_i\| \cdot \|\mathbf{p}_i\|}$$

Can We Find Simultaneous Discoveries?



EDM Successfully Identifies Scientific Breakthroughs



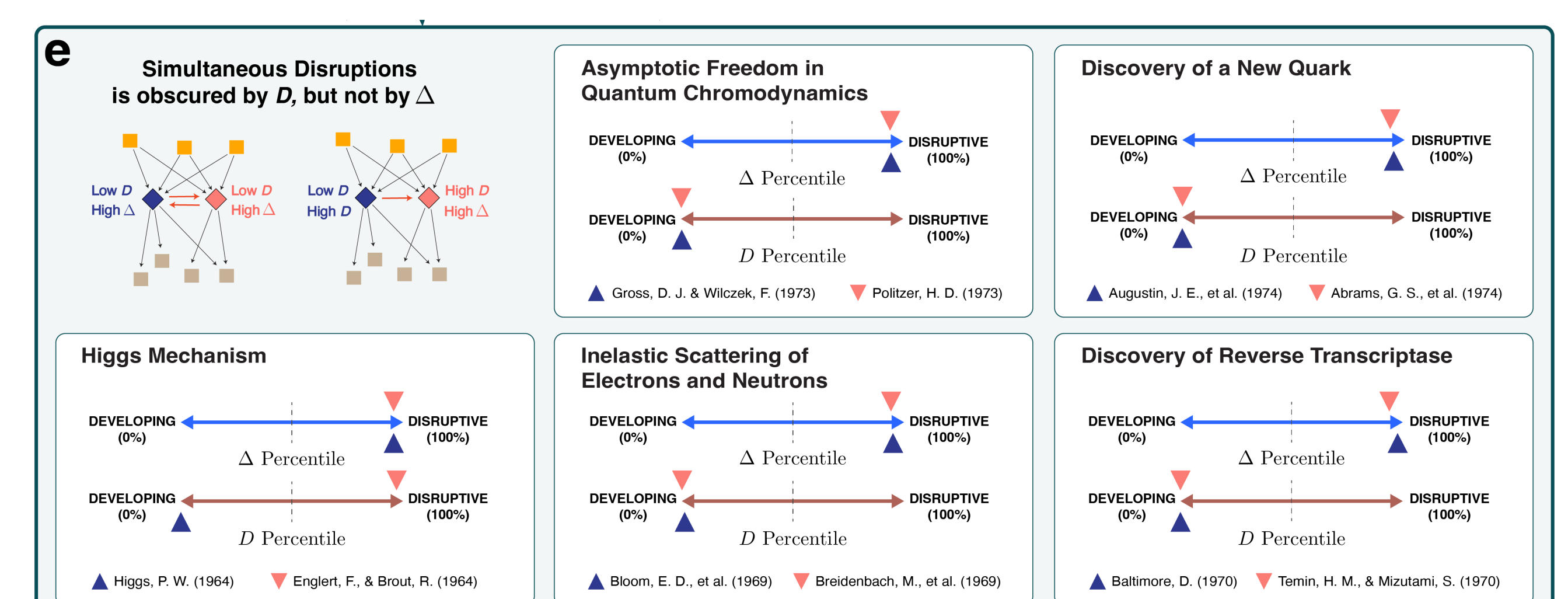
55M+ papers (Web of Science) + 644K papers (APS)

Better identification of landmark papers: Δ ranks most Nobel Prize-winning and milestone papers as highly disruptive, eliminating the bimodal distribution seen with D

Robust beyond citation counts: Randomized networks preserving citation counts, reference counts, and citation year gaps yield different Δ distributions.

Statistically significant predictor: Firth's logistic regression shows Δ has significant odds ratios (OR = 1.34 for Nobel, 1.23 for milestone, $p < 0.001$), while D shows no significance ($p > 0.05$)

Simultaneous Disruptions Uncovered



Papers showing the largest gap between EDM and D are mostly simultaneous discoveries. When simultaneous discovery cite each other, that mutual citation distorts D — flipping it from one extreme to the other.

Identifying Simultaneous Discovery via Future Vector Proximity

- Papers representing the same discovery cluster as nearest neighbors in embedding space
- Identified 18,417 potential simultaneous discovery pairs in APS dataset
- Among 80 high-impact pairs (>300 citations), 80% confirmed as simultaneous discoveries

Resources

Code: github.com/yy/embedding-disruptiveness

Blog: <https://munjungkim.github.io/embedding-disruptiveness-blog/>

Package: [embedding-disruptiveness \(PyPI\)](#)



Code



Blog



Package